

Session 13 – Advanced Lab Sheet

Dataset: Student Performance (Kaggle dataset)

Goal: Predict *grade* (A/B/C/D/F) and evaluate classification performance.

PART A — Load and Explore the Dataset

Q1. Import required libraries.

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, roc_auc_score, classification_report
from sklearn.ensemble import RandomForestClassifier
import matplotlib.pyplot as plt
import seaborn as sns
```

Q2. Load the dataset.

```
# Define the file path (the CSV stored in Google Drive)
file_path = "/content/drive/MyDrive/Datasets/Session11/student_performance.csv"

# Load the dataset
df = pd.read_csv(file_path)

# Display the first few rows
print(df.head())

# Display info about the dataset
print(df.info())
```

Q3. Check missing values

PART B — Convert Grades into Pass/Fail and Prepare Features

Q4. Convert grade → binary classification (Pass/Fail).

- Pass → A or B
- Fail → C, D, F

Q5. Select features and target.

Q6. Train–test split

Q7. Scale numeric features

PART C — Train a Model (Random Forest)

Q8. Train Random Forest model

Q9. Predictions

PART D — Evaluate Using All Classification Metrics

Q10. Accuracy

Q11. Precision & Recall & F1

Q12. Confusion Matrix

Q13. ROC & AUC

PART E —Understanding Questions

These are advanced questions to test their analysis skills.

Q14. Even if accuracy is high, what can go wrong?

Q15. If precision is high but recall is low — what does it mean?

Q16. If recall is high but precision is low — what does it mean?

Q17. Why is AUC important for evaluating models?

Q18. Based on confusion matrix, does the model overpredict PASS or FAIL?

Q19. Which feature seems most important for prediction? Why?

Q20. Is the dataset imbalanced? If yes, how does it affect metrics?